Gavin Lynch

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EDUCATION

California Polytechnic State University — **San Luis Obispo, CA**

MS in Computer Science (GPA: 4.0) — *Graduation Date: Aug 2026*

- Research Focus: Applying machine learning to genetics and multiomics analysis

BS in Computer Science, AI/ML Concentration (GPA: 3.6) — *Graduation Date: June 2025*

WORK EXPERIENCE

Wavepoint Health (Central Coast Neurology) — **Remote**

Software Engineer — *June 2026 – Current*

- Architected and shipped full-stack features across a **Next.js/GraphQL/PostgreSQL/AWS** stack, from schema design and API resolvers to production UI, for a HIPAA-regulated clinical platform.
- Designed and built a configurable rules engine in **TypeScript** that parses YAML-defined domain configs to drive evaluation logic, replacing hardcoded business rules with a data-driven system.
- Integrated 5+ third-party REST/OAuth APIs (Dexcom, Oura, Strava, InBody) and led infrastructure/security hardening (AWS CDK, Secrets Manager) to meet HIPAA compliance requirements.

Novo Nordisk — **Raleigh, NC**

Data Engineer Intern — *May 2025 – July 2025*

- Engineered a dynamic error reporting tool using **Python** and **Pandas** that automated the detection of system errors, reducing manual review time and improving data accuracy for key metrics.
- Utilized **SQL** to manage, extract, and analyze data from large-scale relational databases, improving error management and streamlining reporting processes.
- Redesigned and deployed critical workflows in Syncade MES, integrating new data pipelines that automated manual steps and enhanced overall manufacturing efficiency.

Cal Poly Computer Science Department — **San Luis Obispo, CA**

Teaching Assistant - Intro to Software Engineering — *Sept 2024 – June 2025*

- Guided 90+ students in applying Agile methodologies and **Git/Github** version control, leading to high proficiency demonstrated by the successful, collaborative development of complex semester-long projects.
- Created standardized grading rubrics to evaluate 200+ assignments, improving feedback consistency, reducing grading time, and clarifying key learning objectives for students.

PROJECT EXPERIENCE

Fast-Select — *Jun 2025 – Current*

- Developed fast-select, a high-performance Python library using **Numba/CUDA** to accelerate feature selection algorithms by up to 50-100x, overcoming computational bottlenecks for large-scale bioinformatics datasets.
- Engineered a professional software package featuring a Scikit-Learn compatible API, dual CPU/GPU execution backends for hardware flexibility, and a full **CI/CD** pipeline for automated testing and deployment to PyPI.

RESEARCH EXPERIENCE

Cal Poly Animal Science Department — **San Luis Obispo, CA**

Machine Learning Researcher — *Feb 2024 – Aug 2026*

- Built a custom **PyTorch** Transformer for high-dimensional genomic classification (624k features), outperforming industry baselines by 37–62% on AUPRC via a nested cross-validation pipeline with integrated feature selection.
- Created custom binary compression, alignment, and quality control software to streamline the processing of large-scale genomic datasets, reducing storage footprint and I/O loading times by over 50%.
- Selected to present novel deep learning architectures for genomic selection at the Plant & Animal Genome Conference (PAG 33); awarded 2nd Place at the Cal Poly Animal Science Research Symposium (2024 - 2026).

SKILLS & INTERESTS

Skills: TypeScript | React/Next.js | Node.js | GraphQL | PostgreSQL | Python | AWS | REST/OAuth | CI/CD | Git

Interests: Machine Learning, Software Engineering, Genetics, Agentic AI, Research, Trail Running, Skiing